Water Treatment Technology (CLPE16)

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Lecture 13 (5 August 2025)

Lecture 14 - 15 (8 August 2025)

Discussion on:

- Introduction to physical/chemical water treatment Process

6. Overview method used for water treatment

Overview of Method Used to Treat Water

‘Means’ method used for the treatment of Water

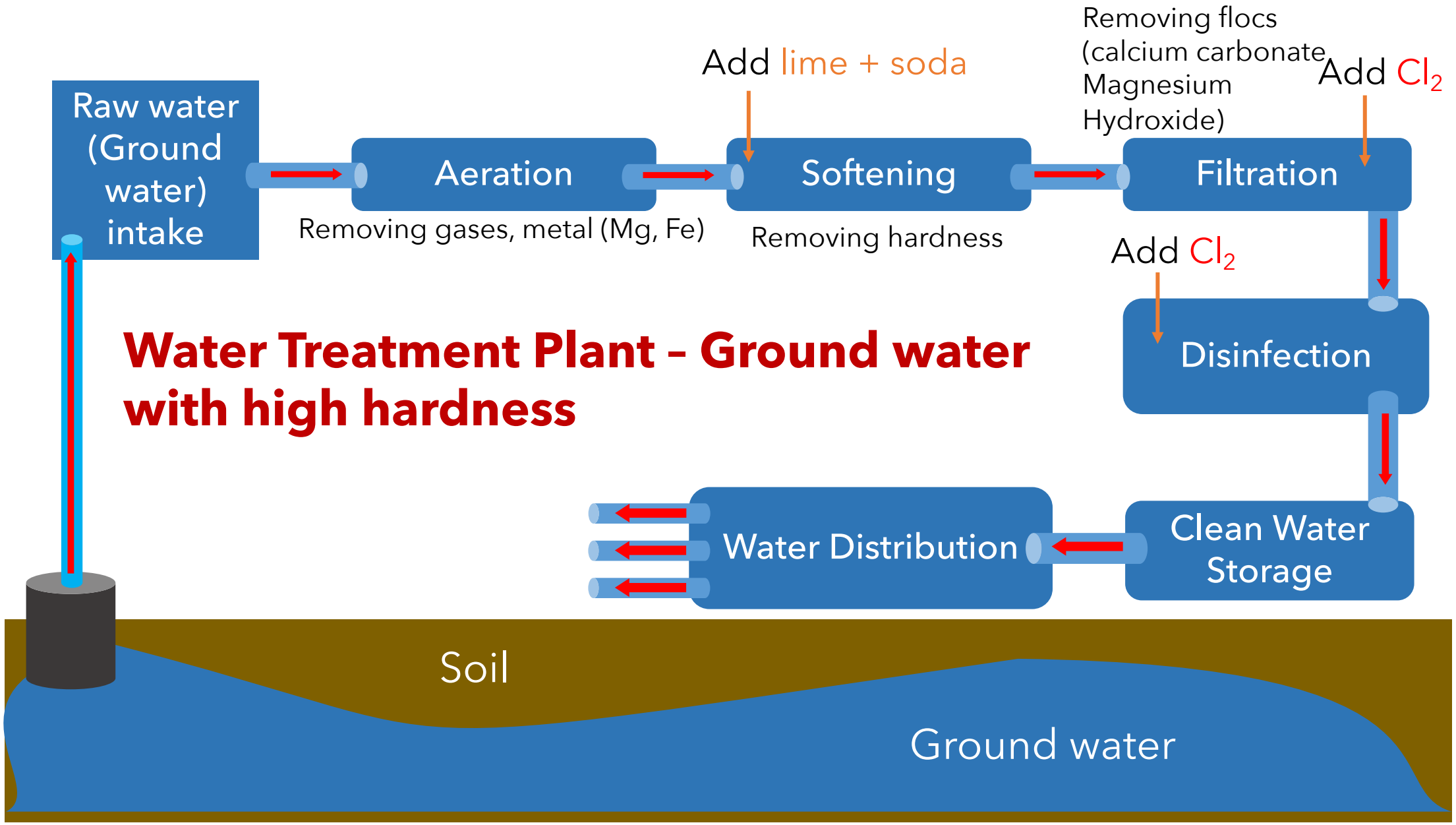
Method Used for the Treatment of Water

- Depending on the types of water, quality of water, and level of contamination - we have to provide the various treatment.

- We have discussed about the various source of water available -surface water or ground water

- Quality of water and level of contamination: – based on physical, chemical, biological parameter

Water Treatment Plant – Ground water with high hardness



Description about the groundwater flow chart:

It is clear from the previous slide flow chart. If source of water is ground water with high hardness. What types of treatment system we have to provide?

- Take the raw water that is ground water then first go for the Aeration, and followed by softening (here add lime as well as soda).
- After softening, we can go for the filtration.
- After filtration, we can go for the disinfection. Then store the clean water and supply (water distribution, here add Cl_2)

This is typical treatment adopted for ground water source with high hardness.

Complete description about the groundwater flow chart:

- First, What we have to do is → We have to tale the raw water then aerate it; The aeration process is for removing undesirable gases present in the water as well as to remove metal like Manganese (Mg), Iron (Fe) ion etc.

- Then If hardness is very high → We have to remove the hardness because excess hardness is not permitted.

- So, How to remove the hardness? For that we have to go for softening process.

- Here, What we are doing is – We are adding line and soda. So, the hardness is coarse because of divalent cations. Especially the presence of calcium (Ca^{2+}) and Magnesium (Mg^{2+}) cause the Hardness.

- Q How can we remove this hardness?

- We can precipitate out this calcium (Ca^{2+}) and Magnesium (Mg^{2+})

- Calcium (Ca^{2+}) can be precipitated out as calcium carbonate. Which is highly insoluble.

$$Ca^{2+} \rightarrow CaCO_3 \downarrow (Calcium\ Carbonate)$$

- And Magnesium (Mg^{2+}) can be precipitated out as Magnesium hydroxide.

$$Mg^{2+} \rightarrow Mg(OH)_2 \downarrow$$

- So, by the addition of this lime and soda.

$$Lime \rightarrow Calcium\ Oxide\ (CaO)$$

$$Lime \rightarrow CaO \xrightarrow[\text{with water}]{\text{Mixed}} Ca(OH)_2$$

$$Soda\ ash \rightarrow Na_2CO_3$$

- When we added Lime and soda ash, whatever calcium and magnesium that is present in the system. It will be getting precipitated as calcium carbonate and magnesium hydroxide.

- So, We can remove the sludge by setting and take the sludge to sludge drying bed and we can recover the lime if it is required.

- After the softening What we have to do is → We have to go for filtration.
- Here, we will be using sand filters. So, this will be removing the entire leftover. Flocs of calcium carbonate and magnesium hydroxide.
- We can even add some chlorine (Cl_2) to the filter bed to avoid microbial growth on the filter bed.
- As the filter bed is exposed to the atmosphere – there is a chance of bacterial growth. So we can avoid that one by adding chorine (Cl_2).
- After filtration → We can go for disinfection by the addition of chlorine.
- This is what we are practicing in India.

- Instead of chlorine we can go for either ozone or UV radiation and once this treatment is over → We can send it to the distribution system.

- But if the ground water is not having Hardness, and for most of the ground water. Depending upon the locality and the mineralogy of that area the concentration of calcium and magnesium will be varying.

- So, in this case we can just take the ground water and if iron and manganese not present → then we can go got disinfection and distribution system (that is supply.)

- If ground water is contaminated → Then this treatment may not sufficient → Then we have to go for other additional treatment.

- So, treatment process can not generalized. It all depending upon the type of water available and what is the purpose of water that we are treating for or making for or factor that is the things decide the treatment system.

- For example, we will see if instead of ground water we are using highly contaminated surface water → Then what are the treatment we have to provide.

➢ Typical Treatment Processes For Ground Water

Depending on the quality of the ground water. There are four common treatment trains for treating ground water are identified as follows:

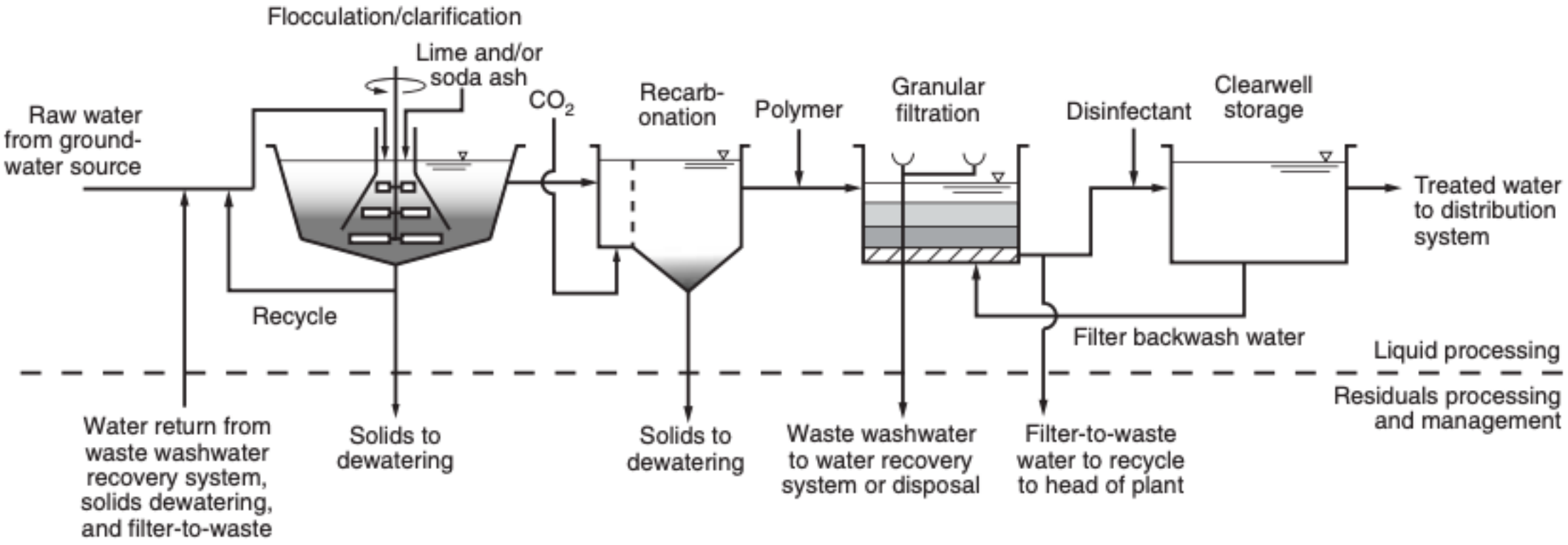
[1] Conventional lime-softening Treatment

[2] Membrane water softening

[3] Gas Removal Treatment

[4] Iron and Manganese treatment

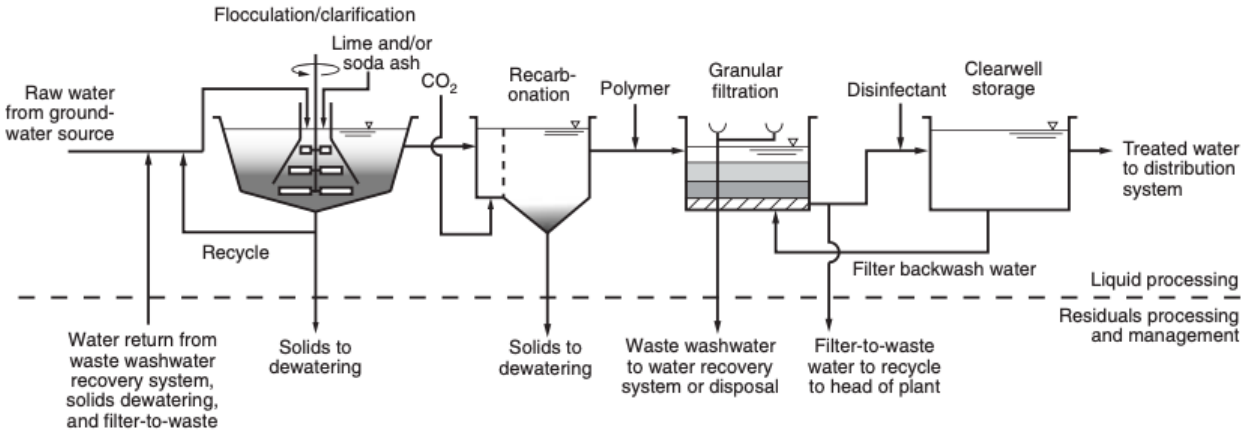
1. Conventional lime-softening Treatment



Treatment of Ground Water by Conventional Lime-Softening Treatment

The diagram illustrates a typical process train for removal of hardness (calcium and magnesium) from groundwater using the lime–soda softening process.

1. Conventional lime-softening Treatment



- **Disinfection:** In general we use chlorine or other disinfectants chemical. It will Kill remaining pathogens and maintain residual disinfectant level for distribution.
- **Clearwell Storage:** Store treated water temporarily. It will provide contact time for disinfection before distribution.
- **Distribution:** Treated water is sent to the distribution system for consumer use.

- **Raw Water Intake:** Source of Water → Ground water, and Initial condition → contains calcium (Ca^{2+}) and magnesium (Mg^{2+}), the primary contributors to water hardness.
- **Flocculation/Clarification:** Add Lime ($Ca(OH)_2$) and/or soda ash (Na_2CO_3). Lime removes carbonates hardness (temporary hardness). Soda ash remove non-carbonate hardness (permanent hardness)
- **Recarbonation:** Adding CO_2 gas. It stabilizes the water **pH** by forming carbonic acid. Also it helps to re-dissolve any excess lime.
- **Polymer Addition:** To enhances coagulation and flocculation.
- **Granular Filtration:** Water passes through granular media (usually sand). It remove remaining suspended solids. By doing, It will improve turbidity and clarity.

- **Flocculation/Clarification:** In the flocculation/clarification stage of the lime-soda softening process, the typical chemical reactions involve the precipitation of calcium and magnesium ions, which are responsible for water hardness.

- Removal of Temporary (Carbonate) Hardness: When lime [$Ca(OH)_2$] is added, it reacts with bicarbonate alkalinity to precipitate calcium carbonate ($CaCO_3$):

$$Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3\downarrow + 2H_2O$$

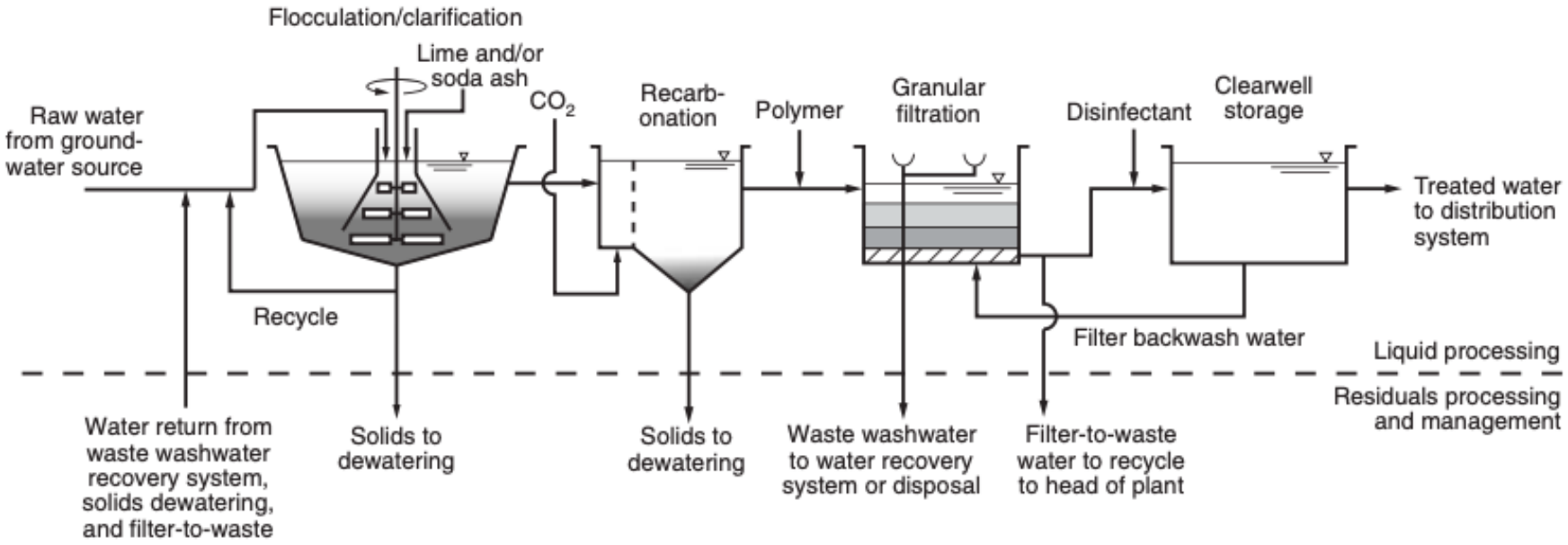
$$Mg(HCO_3)_2 + 2Ca(OH)_2 \rightarrow Mg(OH)_2\downarrow + 2CaCO_3\downarrow + 2H_2O$$

- Removal of Permanent (Non-carbonate) Hardness: For magnesium associated with non-carbonate salts (e.g., magnesium chloride), lime reacts to form magnesium hydroxide:

$$MgCl_2 + Ca(OH)_2 \rightarrow Mg(OH)_2\downarrow + CaCl_2$$

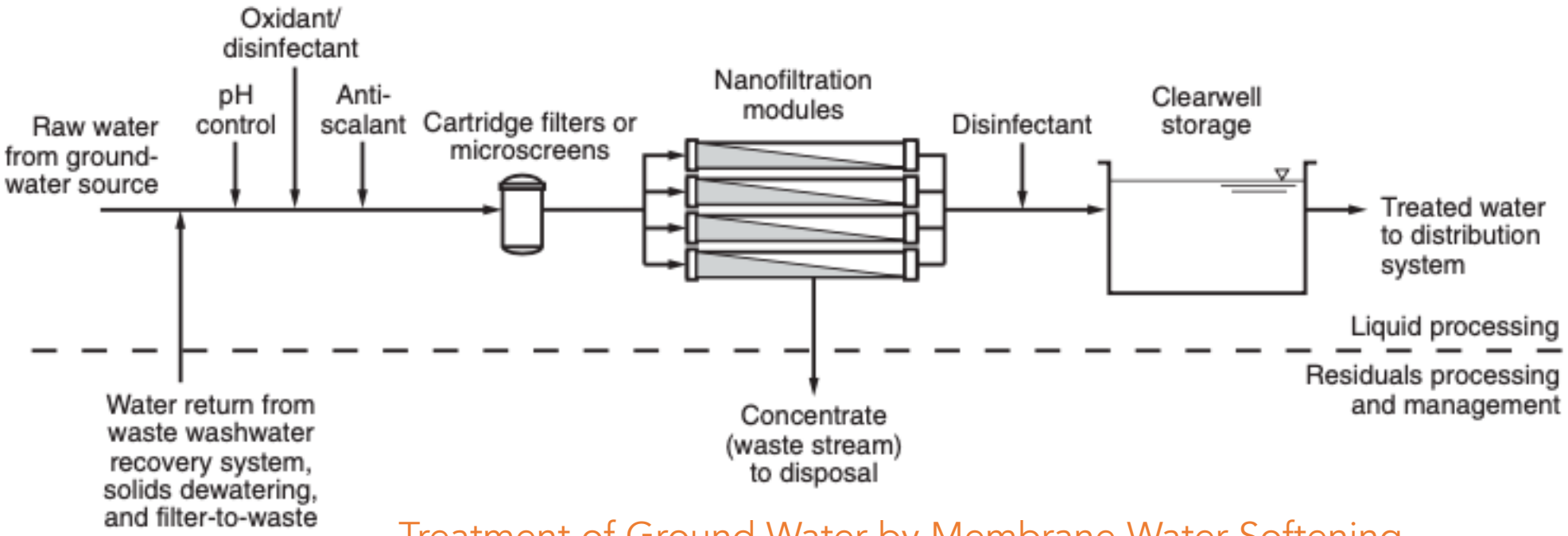
Soda ash (Na_2CO_3) is added to remove calcium associated with non-carbonate hardness, forming calcium carbonate:

$$CaSO_4 + Na_2CO_3 \rightarrow CaCO_3\downarrow + Na_2SO_4$$



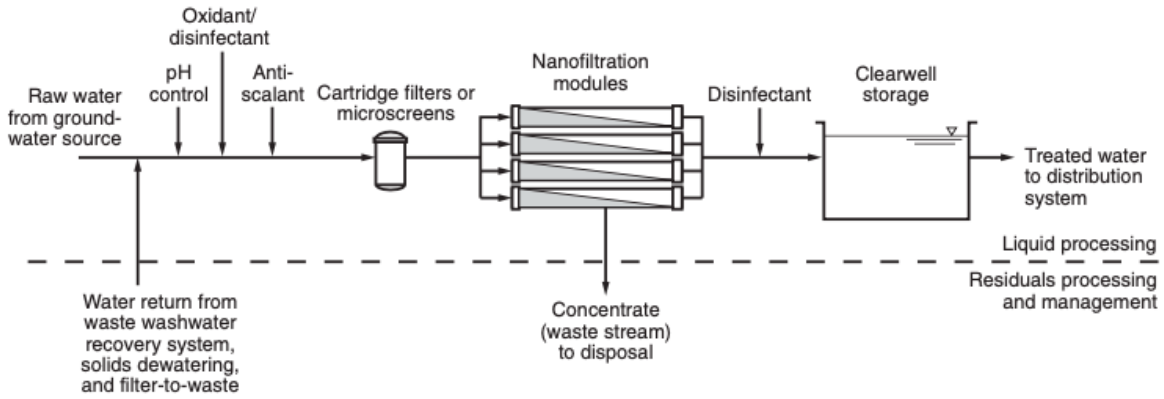
The above diagram illustrates a typical process for removing hardness (calcium and magnesium) from groundwater using the lime–soda softening method. Raw groundwater first enters a flocculation/clarification unit where lime and/or soda ash are added to precipitate calcium as calcium carbonate and magnesium as magnesium hydroxide. The resulting solids settle and are removed for dewatering, while some water is recycled. The clarified water then enters a recarbonation tank where carbon dioxide is added to stabilize pH and remove excess lime. Polymers are added to further aid in coagulation before the water passes through a granular filtration unit to remove any remaining suspended solids. The filters are periodically backwashed, generating waste washwater that is either recycled or sent for disposal. After filtration, a disinfectant is added to kill pathogens, and the water is stored in a clearwell to ensure sufficient contact time before being distributed. Throughout the process, residual solids and waste streams are managed through separate dewatering and recycling systems, ensuring both efficient water treatment and effective waste handling.

2. Membrane water softening



Treatment of Ground Water by Membrane Water Softening

The diagram illustrates a typical process train for groundwater treatment using a nanofiltration softening process.



- **Raw Water Intake:** Groundwater is pumped into the system as the raw water source.

- **Pre-treatment additions:**
  pH Control: Chemicals are added to adjust the pH of the incoming water to optimal levels for nanofiltration performance.
  Oxidant/Disinfectant: Added to kill or deactivate microorganisms and oxidize certain contaminants (e.g., iron, manganese).
  Antiscalant: Prevents scaling on membrane surfaces by inhibiting precipitation of minerals like calcium and magnesium.

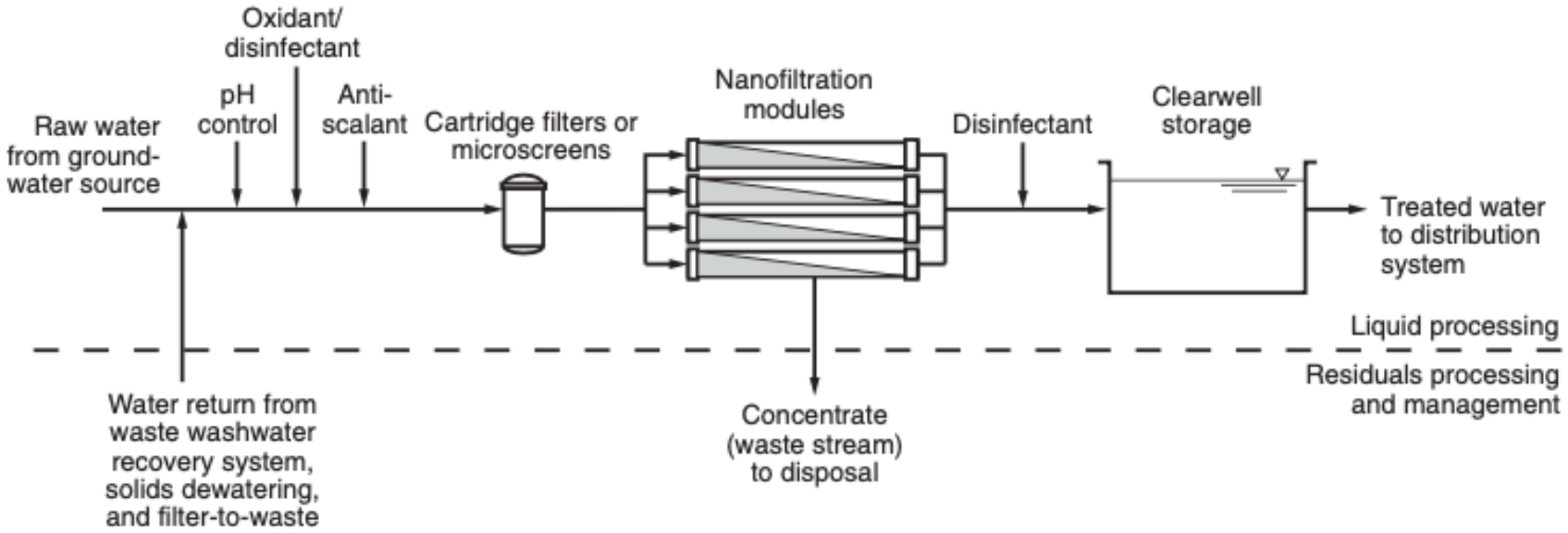
- **Cartridge Filters or Microscreens:** The water then passes through cartridge filters or microscreens to remove fine particulates and protect the nanofiltration membranes from clogging or fouling.

- **Nanofiltration Modules:** The core process where softened water is produced. Nanofiltration membranes remove divalent ions (e.g., Ca^{2+}, Mg^{2+}), organics, and pathogens, while allowing monovalent salts to pass through.

- **Post-treatment Disinfection:** Additional disinfectant is added to the premeate to ensure microbiological safety before storage and distribution.

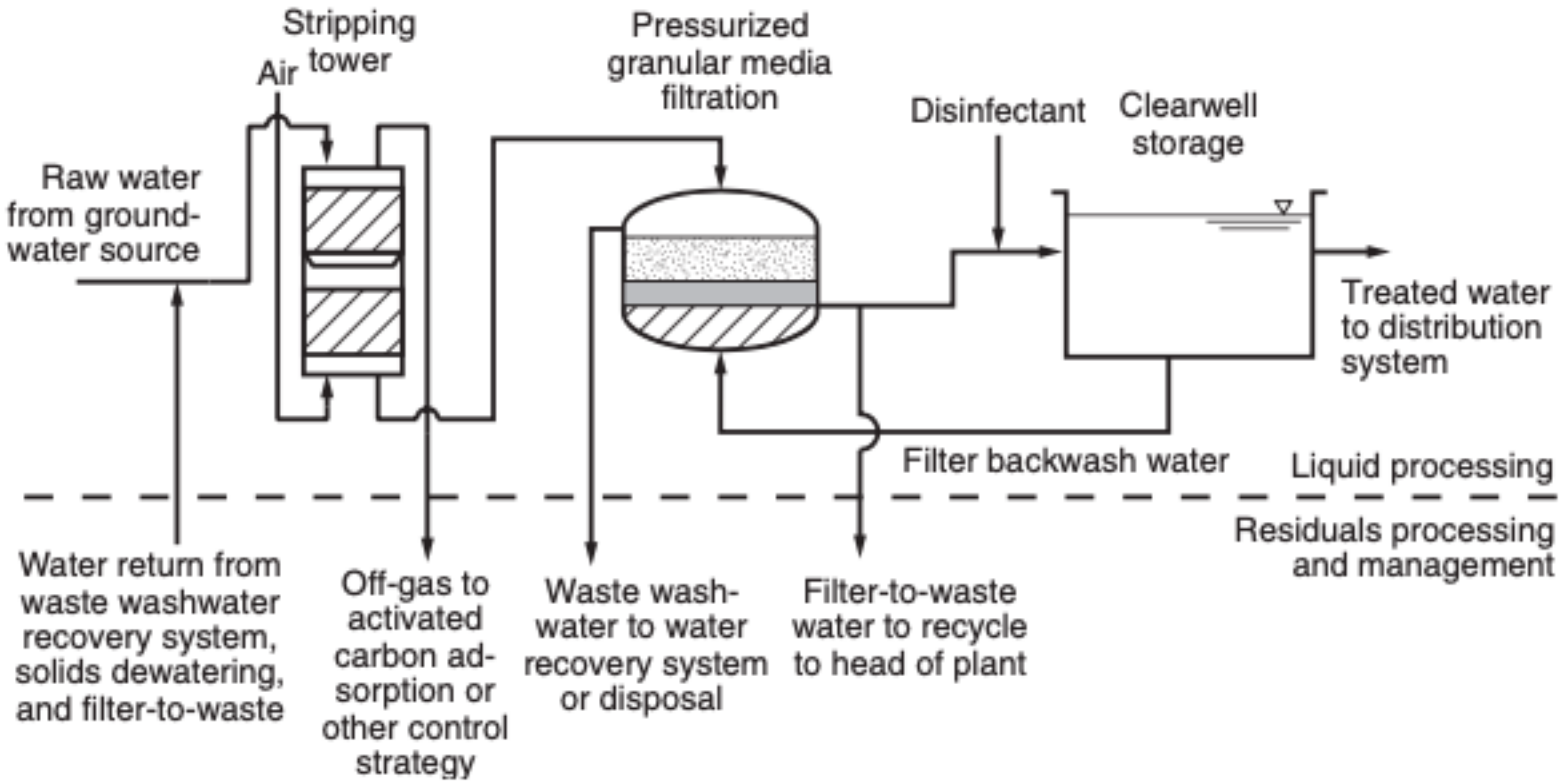
- **Clearwell Storage:** The disinfected treated water is stored in a clearwell tank, providing a buffer and contact time for the disinfectant to work.

- **Distribution:** Treated water is sent to the distribution system for consumer use.



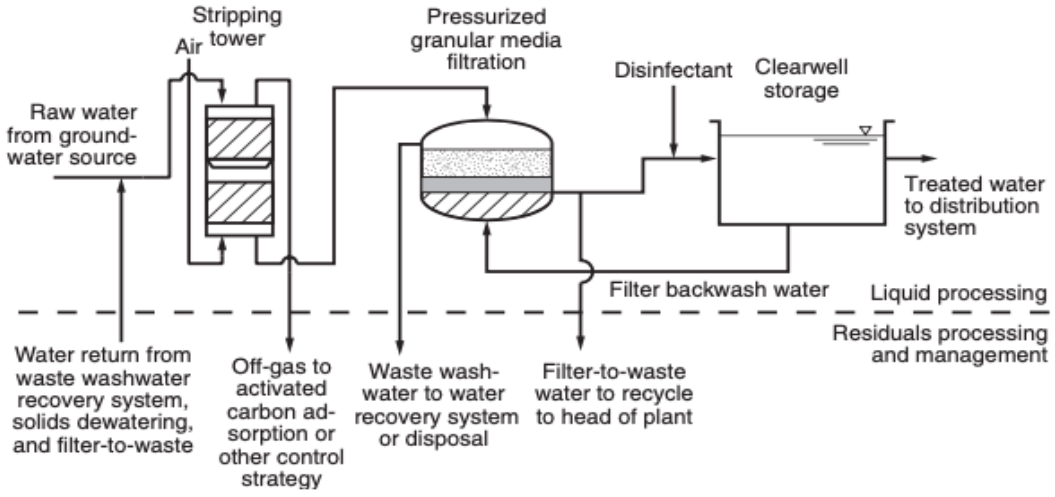
The diagram illustrates a typical process train for treating groundwater using the nanofiltration softening process. The process begins with the intake of raw water from a groundwater source, which undergoes several pre-treatment steps. Chemicals are added for pH control to optimize the conditions for membrane filtration, followed by the addition of oxidants or disinfectants to kill microorganisms and an antiscalant to prevent mineral scaling on membrane surfaces. The pre-treated water then passes through cartridge filters or microscreens to remove fine particulates and protect the downstream nanofiltration membranes. In the nanofiltration modules, the water is separated into two streams: the permeate (softened and treated water) and the concentrate (waste stream). The permeate proceeds for further disinfection to ensure microbiological safety and is then stored in a clearwell storage tank, allowing sufficient contact time for disinfectants before being released into the distribution system. Meanwhile, the concentrate is directed to disposal, and any waste streams from processes like filter backwash, solids dewatering, or wastewater recovery can be returned to the initial stages of the system. The process concludes with residuals processing and management to handle the waste products generated, ensuring both efficient water treatment and responsible waste handling.

3. Gas Removal Treatment



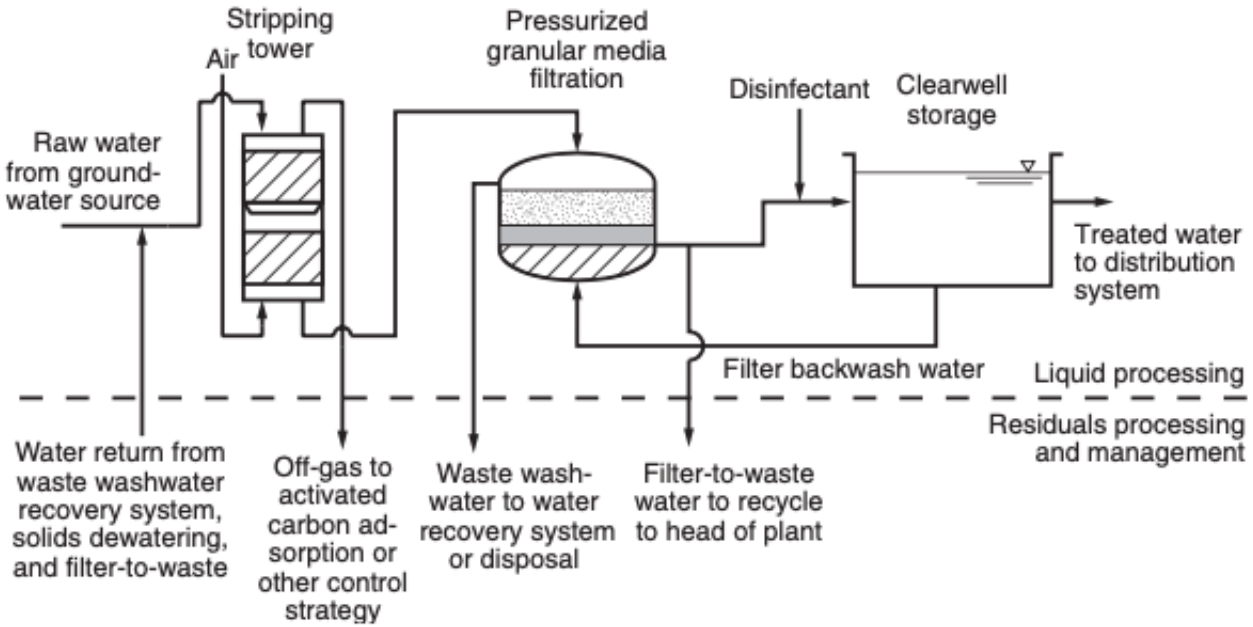
Treatment of Ground Water by Gas Removal Treatment

Gas Removal Treatment



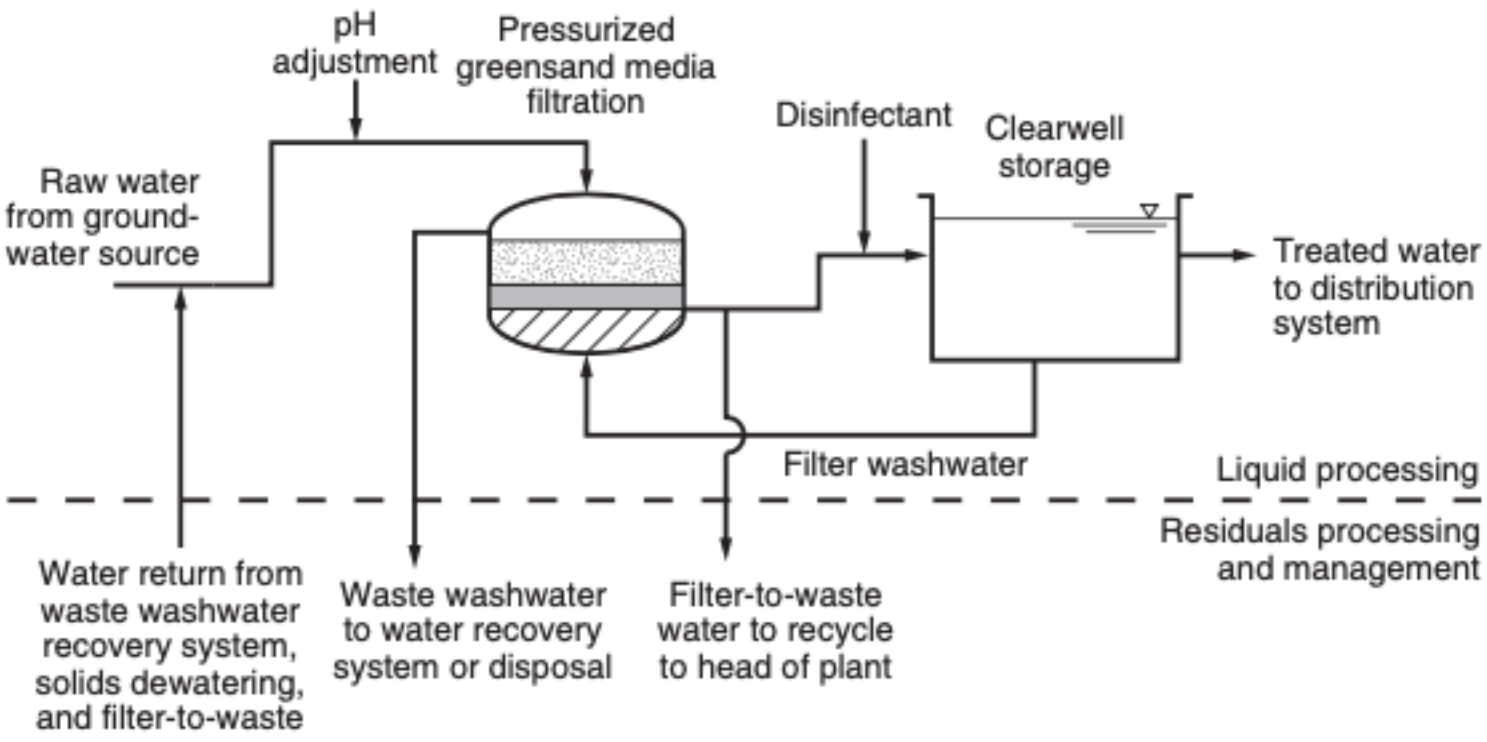
- **Disinfection:** A disinfectant (such as chlorine) is added to eliminate any remaining pathogens, ensuring microbiological safety.
- **Clearwell Storage:** The treated and disinfected water is stored in a clearwell, which provides contact time for the disinfectant and acts as a buffer before distribution.
- **Treated Water Distribution:** The final treated water is then sent to the distribution system for consumer use.

- **Raw Water Intake:** Groundwater containing dissolved gases (e.g., CO_2, CH_4, H_2S) or volatile organic compounds (VOCs) is pumped into the treatment system
- **Stripping Tower (Air Stripping):** Raw water enters a stripping tower, where it flows downward through packing material. Air is blown upward through the tower, promoting mass transfer of volatile/dissolved gases from water to air. This step removes gases and VOCs from the water via air stripping.
- **Off-gas Treatment:** The air leaving the stripping tower, now carrying the removed gases, is treated via activated carbon adsorption or another air pollution control strategy to avoid releasing harmful substances into the atmosphere.
- **Pressurized Granular Media Filtration:** After stripping, the water passes through a granular media filter (e.g., sand, anthracite) to remove any particulates or residual contaminants. Filters operate under pressure to enhance flow and treatment efficiency.



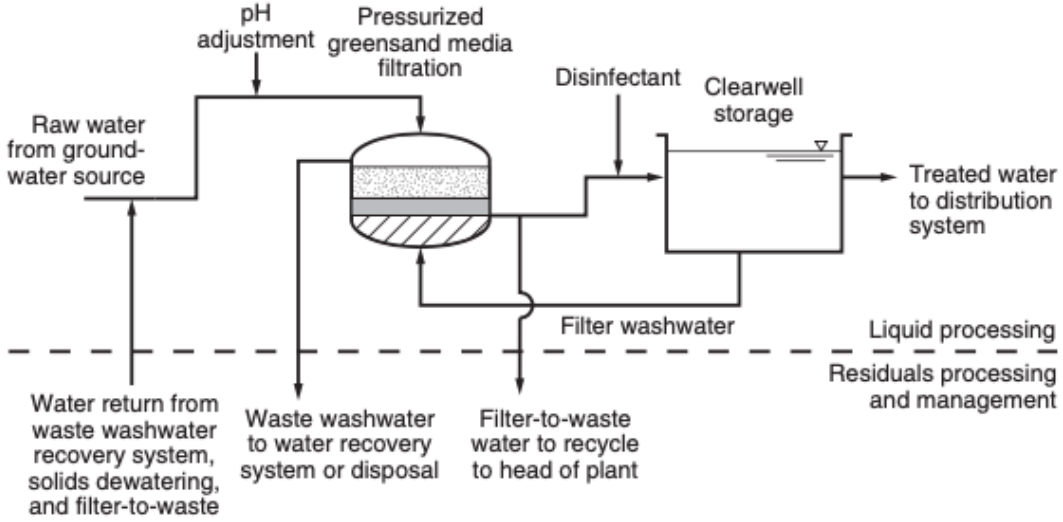
The gas removal treatment process depicted in the diagram is designed to eliminate dissolved gases and volatile constituents from groundwater to produce safe, treated water. Raw water from a groundwater source first enters a stripping tower, where it flows downward through packing material while air is blown upward. This counter-current air stripping facilitates the transfer of dissolved gases (like CO_2, CH_4, or VOCs) from the water to the air. The contaminated off-gas is then treated using activated carbon adsorption or other control methods to prevent atmospheric pollution. The stripped water moves to a pressurized granular media filtration unit, which removes remaining particulates and impurities. A disinfectant is then added before the water enters the clearwell storage tank, where it undergoes final disinfection and is stored temporarily before distribution. Residuals such as filter backwash water and filter-to-waste streams are directed to wastewater recovery systems, with options to recycle treated waste back to the process head. This integrated system ensures the efficient removal of contaminants while managing waste and conserving water.

4. Iron and Manganese Treatment

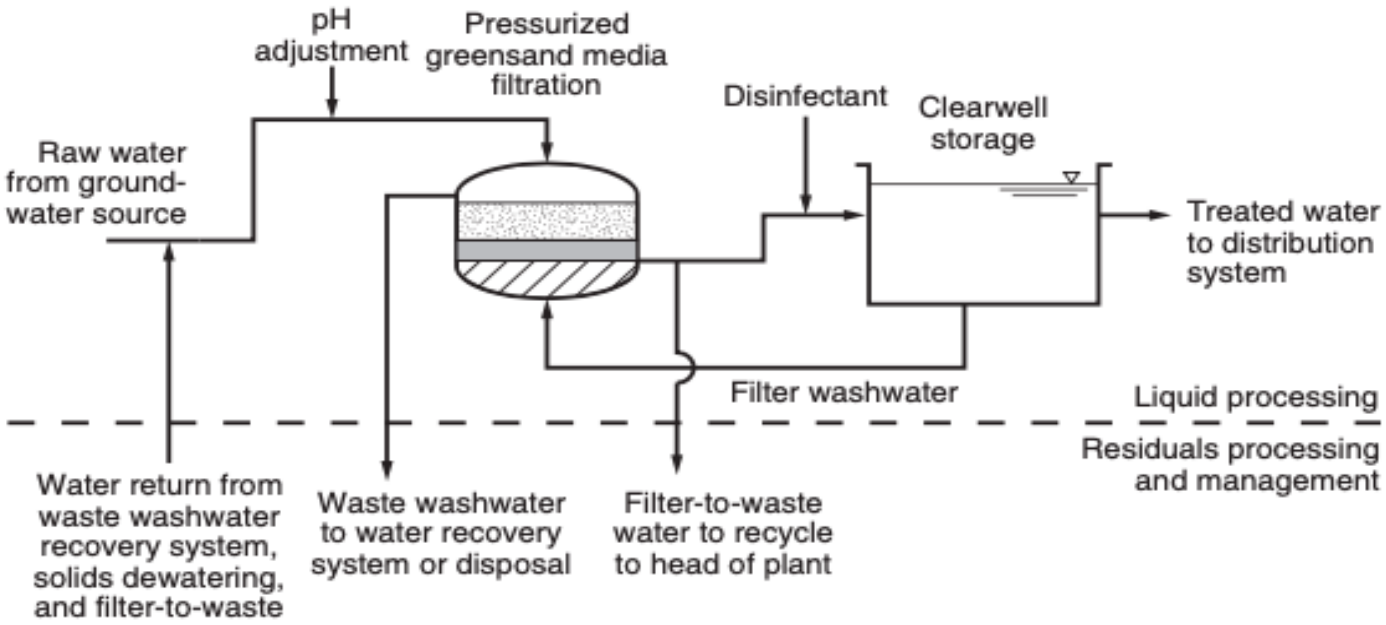


Treatment of Ground Water by Iron and Manganese Treatment

4. Iron and Manganese Treatment

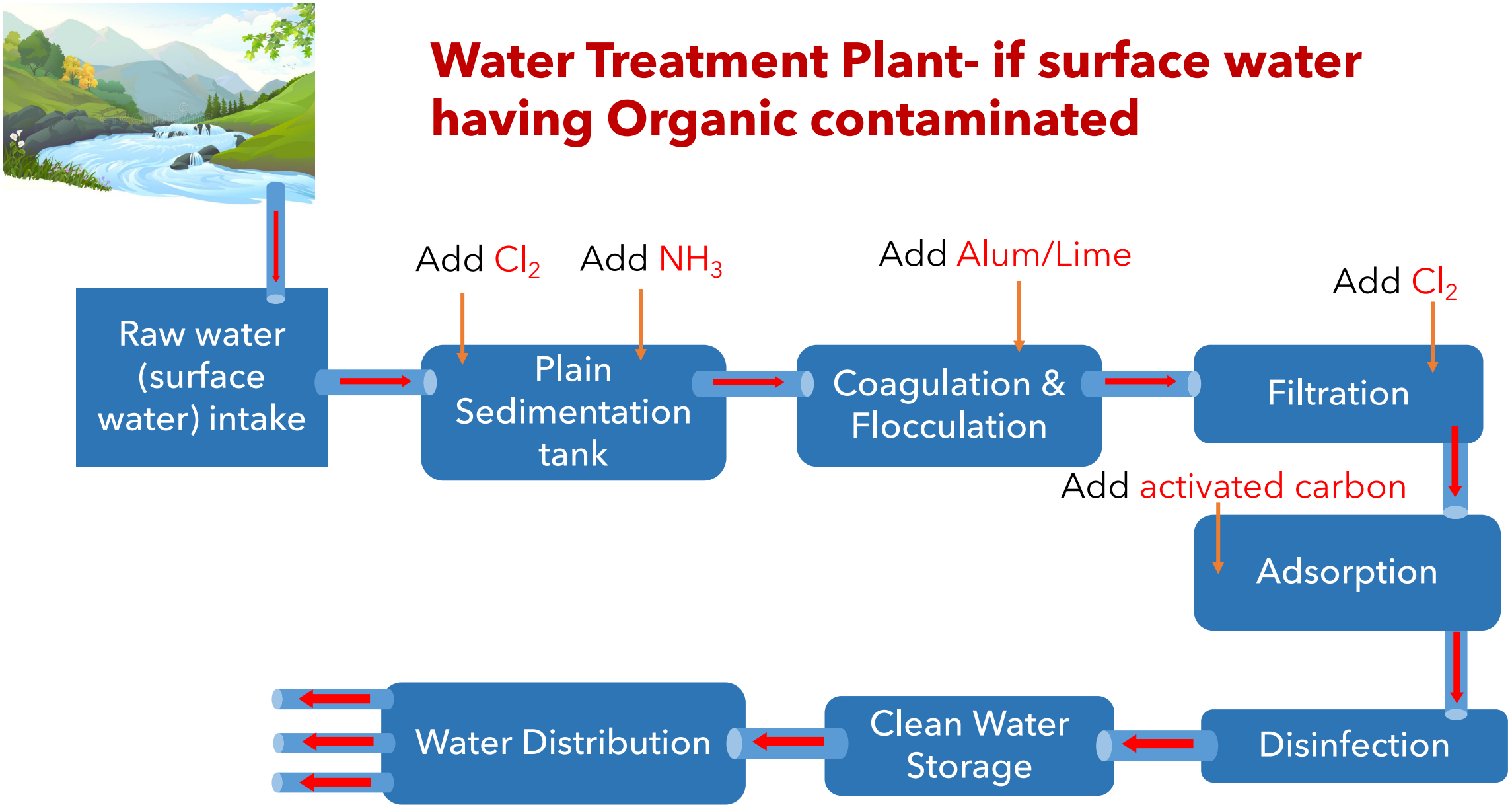


- **Raw Water Intake:** Groundwater containing iron and manganese is pumped into the system.
- **pH Adjustment:** The pH of the raw water is adjusted to optimal levels to promote the oxidation of iron and manganese, which aids in their removal during filtration.
- **Pressurized Greensand Media Filtration:** The pretreated water is passed through a pressurized greensand filter. Greensand is coated with manganese oxide, which acts as a catalyst to oxidize soluble iron and manganese into insoluble forms, which are then trapped in the filter media.
- **Disinfection:** A disinfectant (such as chlorine or potassium permanganate) is added post-filtration to ensure microbial safety and to further aid in oxidation if needed.
- **Clearwell Storage:** The filtered and disinfected water is stored in a clearwell (a storage tank) before being pumped to the distribution system for supply.
- **Residual and Washing Handling:** (a) Filter Washing is periodically generated during backwashing of the greensand filters to remove accumulated iron and manganese solids. (b) Filter-to-waste water is temporarily diverted back to the head of the plant during filter start-up to avoid passing turbid water into the system. (c) Waste washwater can be either disposed of or sent to a water recovery system where solids are dewatered, and the clarified water may be returned to the process.
- **Recycle Loops:** Some of the filtered water or recovered water is recycled back into the system, reducing waste and improving efficiency.



The diagram illustrates a typical process train for removing iron and manganese from groundwater using pressurized greensand media filtration. Raw water from a groundwater source first undergoes pH adjustment to optimize the oxidation of iron and manganese. The water is then passed through a pressurized greensand filter, where the greensand media—coated with manganese oxide—oxidizes and filters out the iron and manganese as insoluble particles. After filtration, a disinfectant is added to ensure microbial safety. The treated water is stored in a clear well before being sent to the distribution system. Periodically, the filter undergoes backwashing to remove accumulated solids, producing filter washwater and waste washwater. These waste streams may be disposed of or sent to a water recovery system for solids removal and possible recycling. Some filtered water may also be returned to the head of the plant as filter-to-waste water, enhancing system efficiency and minimizing water loss.

Water Treatment Plant- if surface water having Organic contaminated



Description about the surface water flow chart:

It is clear from the previous slide flow chart. If source of water is surface water with high suspended solids and settable solids. What types of treatment system we have to provide?

- Take the raw water that is surface water then first go for the plain sedimentation, and followed by coagulation and flocculation.
- After coagulation and flocculation, Then we can go for the filtration.
- After filtration, we can go for the disinfection. Then store the clean water and supply (water distribution, here add Cl_2)

This is typical treatment adopted for surface water source with high suspended matter and settable solids.

Complete description about the surface water flow chart:

- Here, the raw water is coming and if it is having lot of settleable solids → then we can go for a plain sedimentation tank → Because of gravity the settleable solids will be settling down and we can add chlorine (Cl2) or other chemicals for oxidizing the organic matter present or preventing any biological action. Sometime, we can even add ammonia to prevent the biological actions.
- Once the plain sedimentation is over → We will be getting settleable solids in the plain sedimentation tank and this we can take to the sand beds for drying and after the plain sedimentation → All the settleable solids are removed and It will be left over with colloidal particles as well as dissolved solid.
- If we want to remove the colloidal solids → We have to go for coagulation and flocculation.
- Here, we have to add coagulants that means Alum and other polymers.
- So, by addition of these chemicals alum, lime and other polymers the colloidal particles will get desatblized and we can settle them easily.
- After this coagulation flocculation → We will be having a secondary sedimentation tank, where all the flocs will be separated.

- After this coagulation flocculation → We have to go for filtration; wither slow zone filter or rapid zone filter.
- Here, we add even chlorine to prevent the biological growth.
- After filtration → We have to go for adsorption.
- Here, we use either activated carbon or other suitable adsorbents to remove organic content present in the water.
- Once the adsorption is over. The water is almost free from the solid both dissolved, colloidal as well as suspended solids then we have to go for disinfection to make water is bacteriologically safe and during disinfection → We will be providing enough chlorine to take care of all the contaminants present in the water as well as some left over chlorine. Which is known as residual chlorine to take care of whatever the contamination is the minimum contamination in the distribution system and once the disinfection is over → We can put the water into the distribution system for supply.
- Therefore, depending upon the water source we have to decide on the treatment.

➢ Typical Treatment Processes For Surface Water

Depending on the quality of the surface water. There are four common treatment trains for treating surface water are identified as follows:

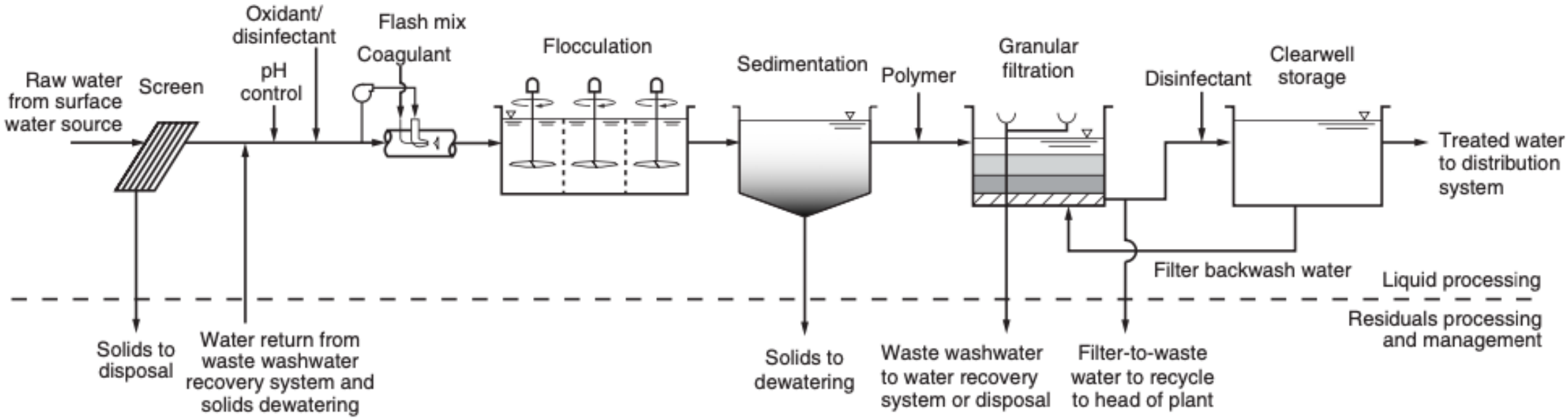
[1] Conventional Water Treatment

[2] Direct- and in-line Filtration Treatment

[3] Membrane Filtration Treatment

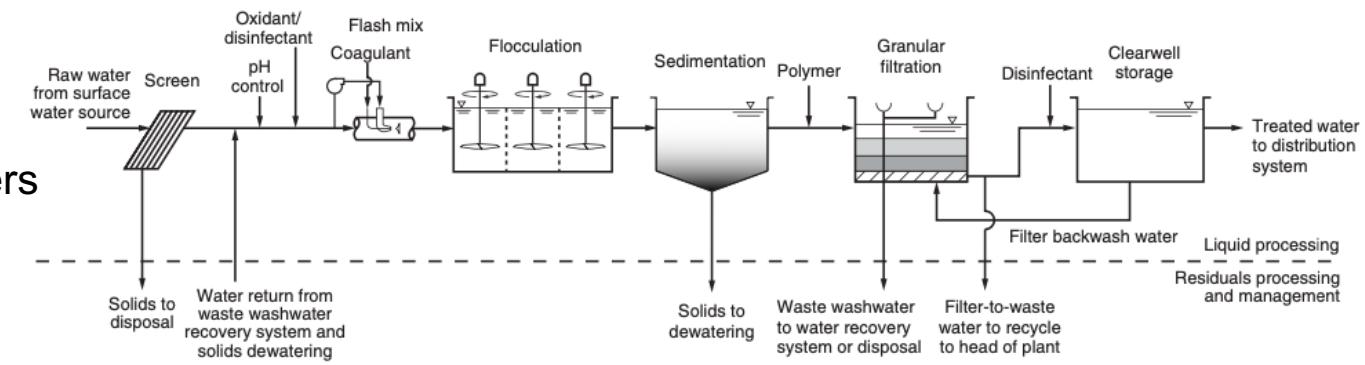
[4] Reverse Osmosis Treatment

1. Conventional Water Treatment

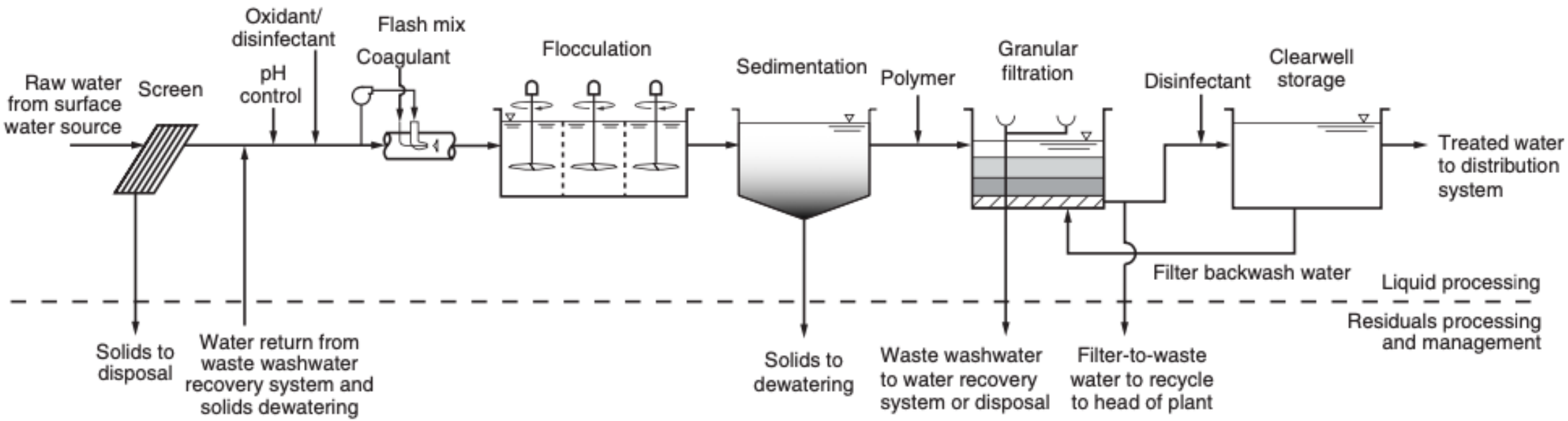


Treatment of Surface Water by Conventional Water Treatment

1. Conventional Water Treatment

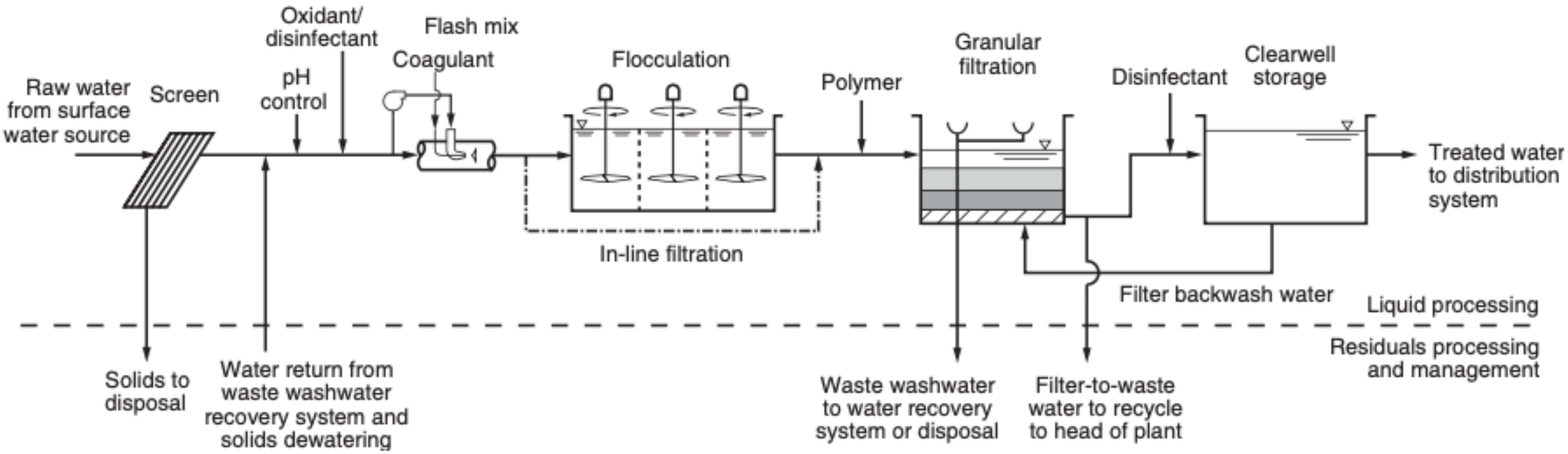


- **Screening**: Raw water from a surface water source enters the plant and passes through screens to remove large debris (e.g., leaves, sticks). These solids are sent to disposal.
- **pH Control and Oxidant/Disinfectant Addition**: Chemicals may be added to adjust the pH and introduce an oxidant or disinfectant (like chlorine) to control biological growth and help with coagulation.
- **Flash Mixing**: A coagulant (such as alum) is rapidly mixed with the water to destabilize suspended particles and form microflocs.
- **Flocculation**: Water flows into flocculation basins, where slow mixing encourages the formation of larger, heavier flocs from the destabilized particles.
- **Sedimentation**: In sedimentation basins, the heavy flocs settle to the bottom as sludge. This sludge is removed and sent for dewatering.
- **Granular Filtration**: The clarified water passes through granular media filters (e.g., sand, gravel) to remove remaining fine particles. A **polymer** may be added before this step to aid filtration.
- **Filter Backwashing**: Periodically, filters are backwashed to clean them. This produces **filter backwash water**, which can be recycled or treated.
- **Clearwell Storage**: Treated water is stored in a clearwell, providing contact time for disinfection and buffering before distribution.
- **Distribution**: The fully treated water is then pumped into the distribution system for consumer use.



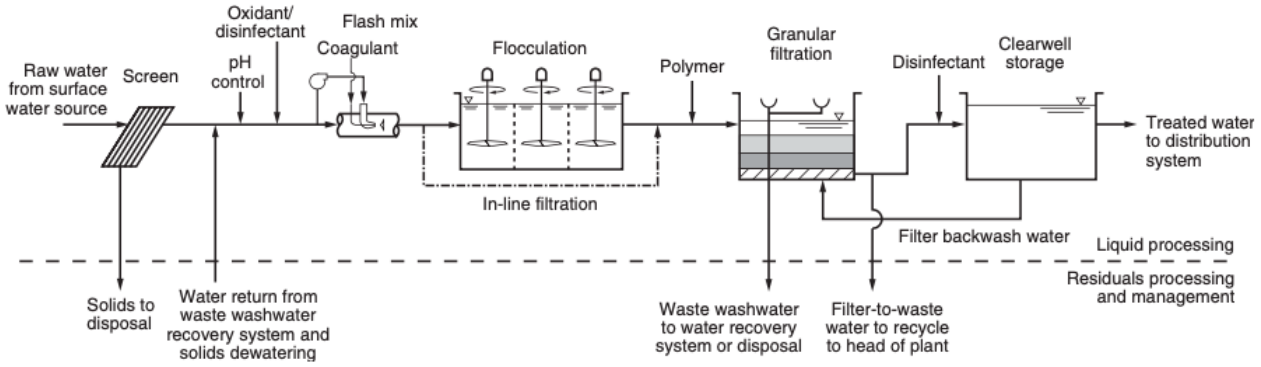
The conventional surface water treatment process involves several sequential steps to ensure the removal of physical, chemical, and biological contaminants. Raw water from a surface source first passes through screens to remove large debris, with the collected solids sent to disposal. pH control and an oxidant or disinfectant are then added to prepare the water for coagulation. In the flash mix chamber, a coagulant is rapidly mixed to destabilize suspended particles, followed by flocculation where gentle stirring forms larger flocs. These flocs settle out in the sedimentation tank, and the resulting sludge is sent for dewatering. The clarified water then undergoes granular media filtration, with optional polymer addition to enhance particle removal. Periodic filter backwashing produces wastewater that is recycled or treated. A final disinfectant is added before the water enters clearwell storage, ensuring sufficient contact time for microbial inactivation. The treated water is then delivered to the distribution system, while all residuals and waste streams are processed and managed separately.

2. Direct and in-line Filtration Water Treatment Process



Treatment of Surface Water by direct and in-line filtration process

2. Direct and in-line Filtration Water Treatment Process



- **Screening**: Raw water from a surface source (like a river or lake) first passes through a **screen** to remove large debris such as leaves, sticks, and other solids. These are sent to disposal.

- **Chemical Addition (pH Control and Oxidant/Disinfectant)**: After screening, chemicals are added to **adjust pH** and apply **oxidants or disinfectants** to control biological activity and aid subsequent treatment steps.

- **Flash Mixing with Coagulant**: A **coagulant** is rapidly mixed into the water in a **flash mix chamber**. This helps destabilize particles and promote the formation of flocs.

- **Flocculation**: The water then enters a **flocculation basin**, where it is gently mixed to allow small particles to clump into larger, settleable flocs. In **in-line filtration**, this step may be bypassed or integrated with filtration depending on design.

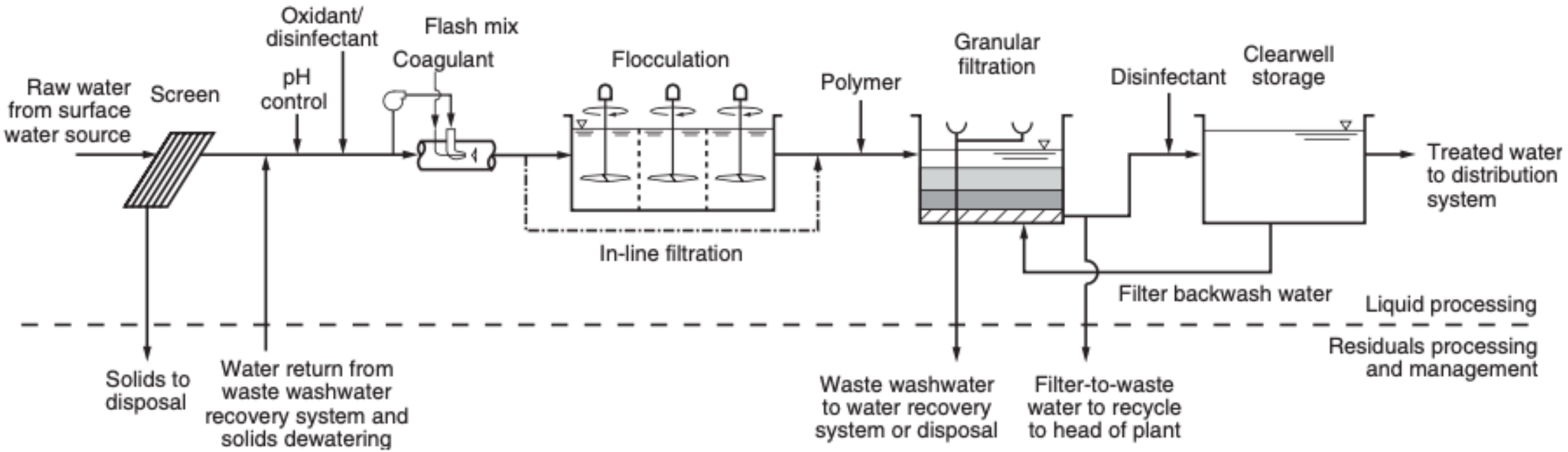
- **Polymer Addition**: A **polymer** may be added before filtration to enhance the removal of fine particles by helping form stronger flocs.

- **Granular Media Filtration**: Water flows through a **granular filtration unit** (e.g., sand or anthracite bed), which physically removes remaining suspended solids.

- **Filter Waste Management**:
  - **Backwash Water**: Used to clean the filter media and is collected for treatment.
  - **Filter-to-Waste**: Initial filtered water after backwashing may be recycled back to the start of the plant until it meets quality standards.
  - **Waste Washwater**: Sent to recovery or disposal.

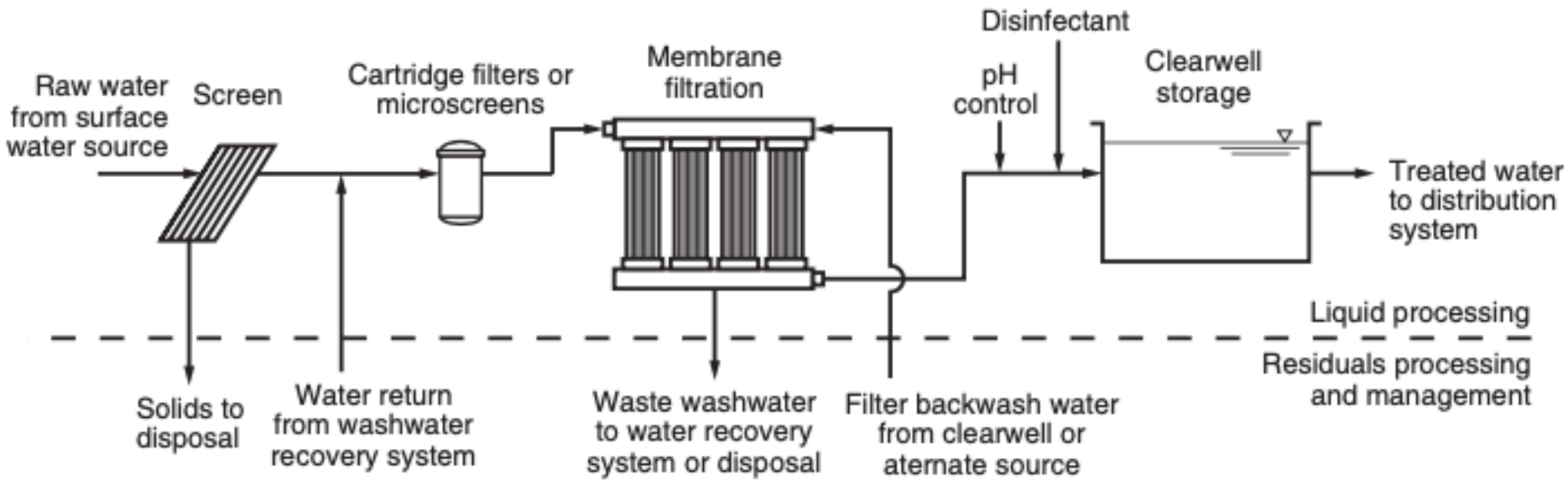
- **Disinfection**: A final **disinfectant** (such as chlorine) is added post-filtration to inactivate any remaining pathogens.

- **Clearwell Storage**: Treated water is stored in a **clearwell**, allowing sufficient contact time for disinfection before distribution.
- **Treated Water Distribution**: The finished, treated water is then sent to the distribution system for public or industrial use.



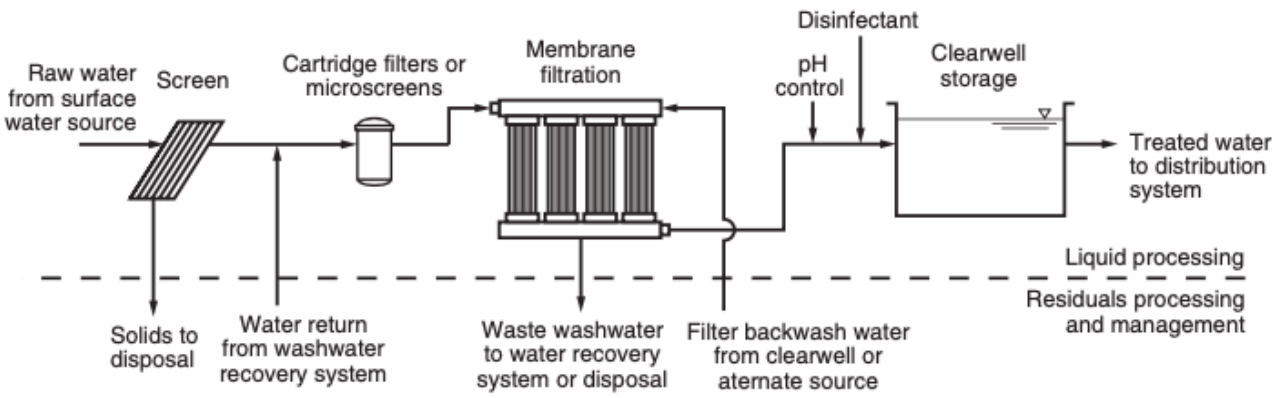
The surface water treatment process using direct and in-line filtration begins with raw water from a surface source passing through a screen to remove large debris, with solids sent to disposal. The water is then treated with chemicals for pH adjustment and the addition of oxidants or disinfectants to control biological activity. A coagulant is rapidly mixed in a flash mixer to destabilize particles, followed by flocculation where particles form larger flocs. In in-line filtration, this flocculation step may be minimized. A polymer may be added to enhance floc formation before the water passes through a granular media filter, which removes remaining suspended solids. Post-filtration, the water is disinfected again and stored in a clearwell to ensure proper disinfection contact time. The treated water is then distributed, while filter backwash water and filter-to-waste water are either recycled or sent for further processing. Residuals like waste washwater and solids are managed through a separate residuals processing system.

3. Membrane Water Treatment Process



Treatment of Surface Water by membrane process

4. Membrane Water Treatment Process



- **Screening**: Raw water from a surface source is first passed through a **screen** to remove large debris and solids (like leaves, twigs, and sediment). The removed solids are sent to disposal.

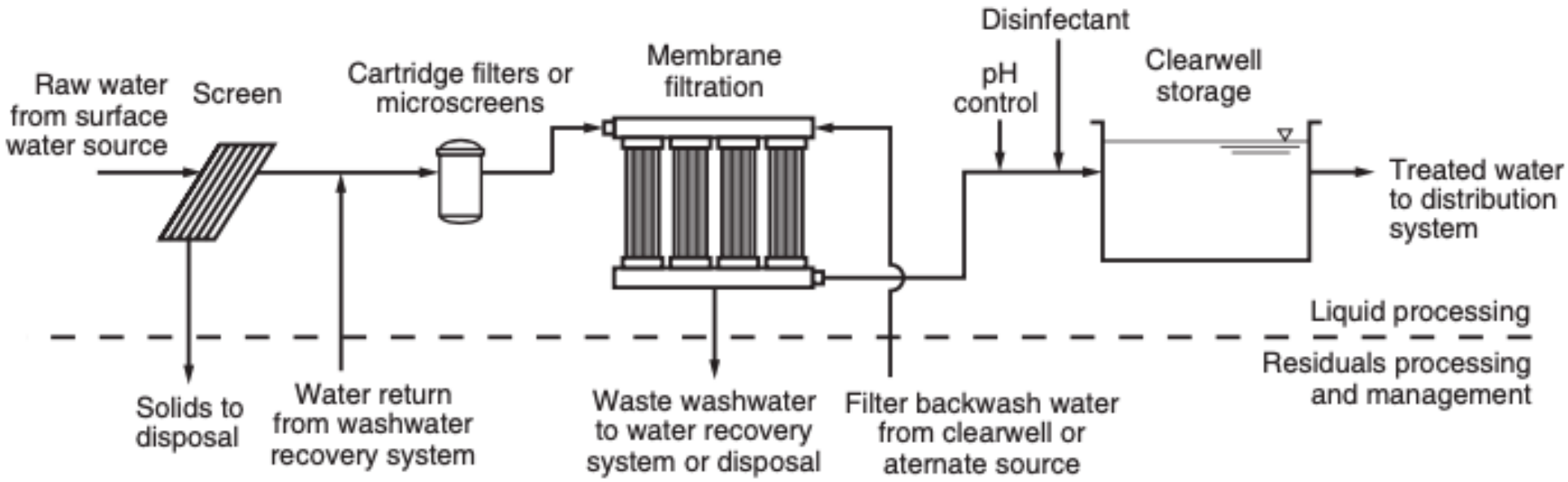
- **Cartridge Filters or Microscreens**: The pre-screened water then passes through **cartridge filters or microscreens** to remove smaller particulate matter. This helps protect the downstream membrane filtration unit from clogging.

- **Membrane Filtration**: In this core step, the water is pushed through a **membrane filtration system** (such as microfiltration, ultrafiltration, or nanofiltration membranes). These membranes physically remove fine particles, microorganisms, and some dissolved contaminants.

- **Chemical Conditioning**: Before the water enters the final storage: (i) **pH control** is done to adjust the water chemistry. (ii) A **disinfectant** (like chlorine) is added to kill any remaining microorganisms and maintain microbial safety.

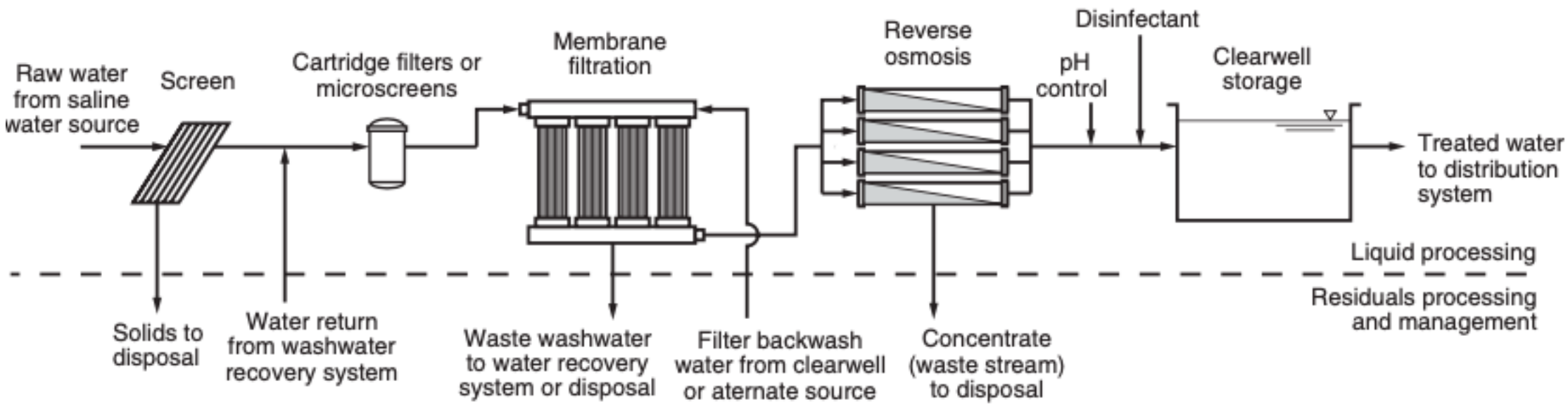
- **Clearwell Storage**: The treated water is then collected in a **clearwell**, a storage tank that provides contact time for disinfection and serves as a buffer for the distribution system.

- **Distribution**: Finally, the clean, treated water is pumped from the clearwell into the **distribution system** for public or industrial use.



The membrane filtration treatment process for surface water begins with raw water passing through a screen to remove large debris and solids, which are sent to disposal. The pre-screened water then flows through cartridge filters or microscreens to eliminate finer particles and protect the membrane unit. In the membrane filtration stage, water is forced through specialized membranes that remove suspended solids, microorganisms, and some dissolved contaminants. Wastewater generated from membrane cleaning is either recovered or disposed of, and backwash water is supplied from a clearwell or another clean source. After membrane filtration, the water undergoes pH adjustment and disinfection to ensure chemical balance and microbial safety. The treated water is then stored in a clearwell to allow for disinfection contact time and is finally distributed for use. Throughout the process, residuals like solids and waste washwater are managed through appropriate disposal or recovery systems to ensure efficient and environmentally responsible operation.

4. Reverse Osmosis Water Treatment Process



Treatment of Surface Water by reverse osmosis process

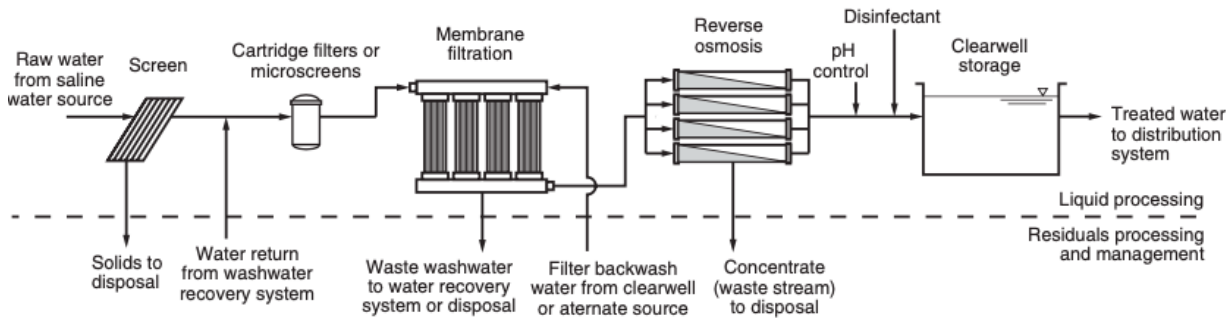
4. Reverse Osmosis Water Treatment Process

- **Screening**: Raw water from a saline source first passes through a screen to remove large debris and solids. The screened solids are sent to disposal.

- **Cartridge Filters or Microscreens**: The pre-screened water then passes through cartridge filters or microscreens for finer removal of particulate matter.

- **Membrane Filtration**: The water undergoes membrane filtration to remove suspended solids and microorganisms. This step generates waste washwater, which is either returned to a washwater recovery system or sent for disposal.

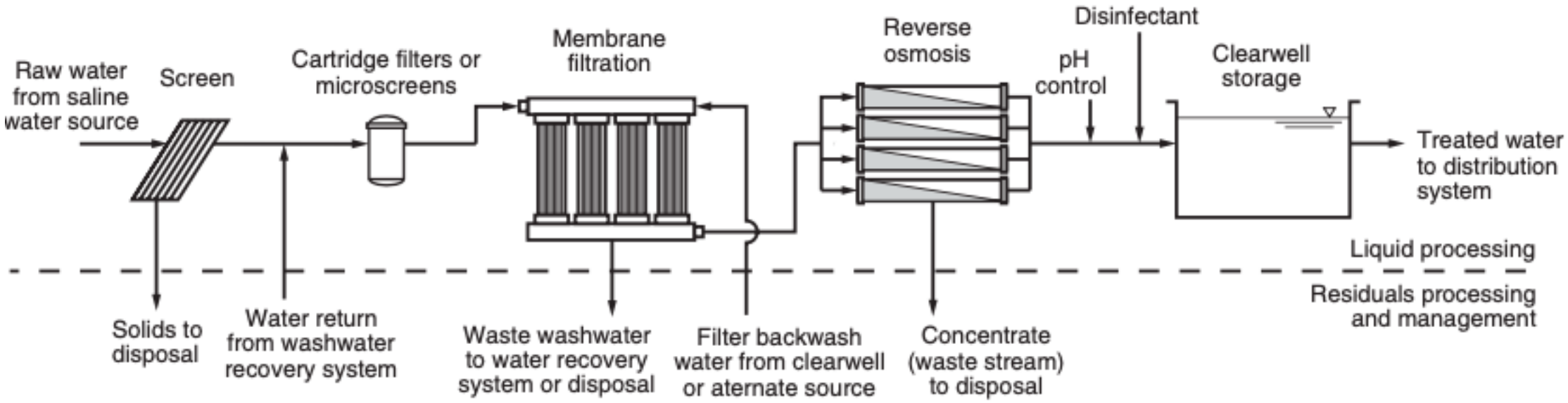
- **Reverse Osmosis (RO)**: In this critical step, high-pressure pumps force water through semi-permeable membranes that remove dissolved salts and impurities. The RO unit produces two streams: (i) **Permeate (treated water)**, which proceeds further in the process. (ii) **Concentrate (brine/waste stream)**, which is sent to disposal.



- **Chemical Conditioning**: **pH Control** and **Disinfection** are applied to the RO-treated water to ensure its safety and stability.

- **Clearwell Storage**: The treated and conditioned water is stored temporarily in a clearwell tank to allow for further mixing and detention.

- **Distribution**: Finally, the treated water is sent to the distribution system for public or industrial use.



The reverse osmosis treatment process for surface saline water begins with screening to remove large solids, which are sent for disposal. The pre-treated water then passes through cartridge filters or microscreens to eliminate finer particles. Next, membrane filtration further purifies the water by removing suspended solids and microorganisms, generating washwater that is either recovered or discarded. The partially treated water enters the reverse osmosis (RO) unit, where it is forced through semi-permeable membranes under pressure to separate dissolved salts and impurities. This produces treated water (permeate) and a concentrated waste stream (brine), which is sent to disposal. The permeate undergoes pH adjustment and disinfection before being stored in a clearwell tank. From there, the clean, treated water is directed to the distribution system, while all waste byproducts are managed through a residuals processing system. This multi-step process ensures high-quality water suitable for various applications.

THANK YOU
Do you have Any Question?